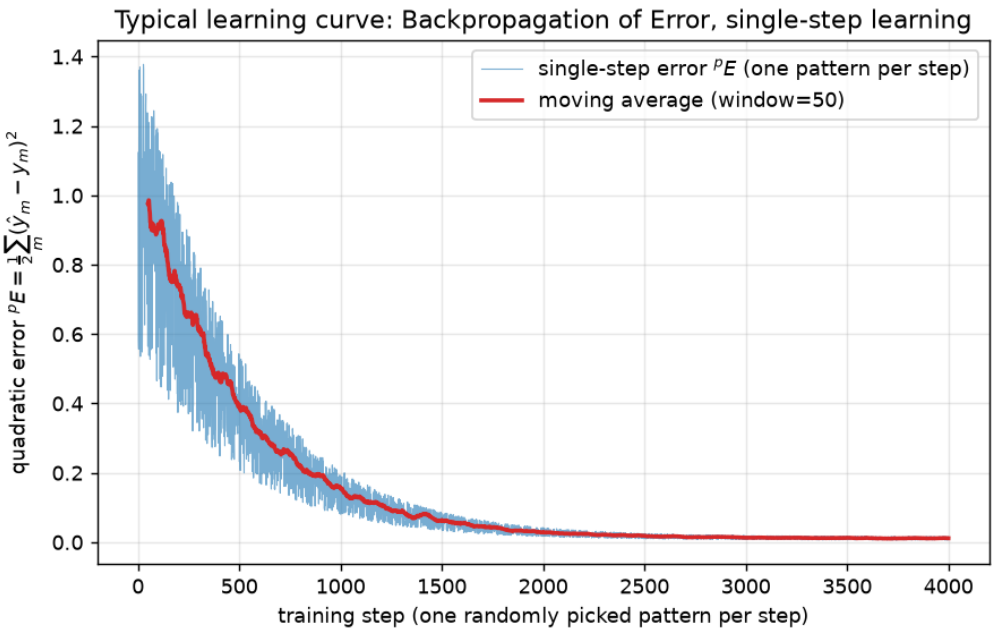
Technical Neural Networks Assignment Sheet 4

Names: Mohamadhosein Sabaghi (50547423)

Assignment 16

With single-step (online) learning, one pattern p is picked and the weights are updated right after it, so the plotted error ${}^{p}E$ is the error of whichever single pattern was just shown – not the total error over all patterns. That makes the curve inherently noisy: it jumps up and down from step to step depending on how well the network currently happens to fit that particular pattern, while only the underlying *trend* decreases smoothly. Typical shape:

- a fast drop at the very start, while the (randomly initialized) weights are still far from any reasonable solution;
- a longer middle phase where the trend keeps decreasing, but individual steps fluctuate a lot around it – "hard" or just-updated-against patterns still produce temporary error spikes;
- a late phase where the trend flattens out near a low floor (ideally near 0, or above 0 if the task isn't perfectly learnable), with the step-to-step noise still visible on top of it – single-step learning never fully smooths out, unlike cumulative/batch learning (assignment 15).



This matches what we actually observed empirically with our own single-step-trained MLPs in PA-A (e.g. `assignments/2/pa_a/data/01/learning_curve.png`): a steep early drop followed by a noisy decreasing trend that settles near a low error floor.